

General Dynamic Model for Skid-Steer Mobile Manipulators With Wheel–Ground Interactions

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Abstract—The design of motion controllers that could reduce terrain disturbances propagated to the arm and optimize the trajectories of skid-steer mobile manipulators (SSMMs) working in agricultural, construction, or mining environments requires accurate physical models capable of correctly representing disturbances caused by changes in terrain slope, loss of traction, and other wheel-ground interactions. Therefore, this paper develops the forward dynamics equations of motion and proposes a general model for SSMMs. The novelty of the model is that it considers a floating-base with nonpermanent wheel-ground contacts, i.e., the mobile base has 6-degree of freedom (DOF) and the wheels may lose contact with the terrain. Unlike the existing models, the proposed one is general because the procedure can be extended to MMs with an arbitrary number of arms or wheels and does not restrict the mobile base to planar 2-D motions. Furthermore, it takes into account the loss of traction and the deformation of the wheel–ground contacts. The experimental results involving a small SSMM with a 3-DOF arm and an industrial skid-steer loader validate the physical fidelity of the model. Although the focus of this paper is not in control aspects, one of the examples shows the role physically accurate models can have in the design of end-effector motion controllers, and its potential application to the design of controllers capable of decoupling the arm motion from ground disturbances.

Index Terms—Dynamics, mobile manipulator (MM), modeling, skid-steer, spatial vector algebra, wheel–terrain interaction.

I. INTRODUCTION

SKID-STEER mobile manipulators (SSMMs) are a common type of robotic platform found in mining excavators (teleoperated or semiautonomous) [1], agricultural service robots [2], [3], military and first responder support robots [4], and planetary

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exploration robots [5] to name a few. While the motion of indoor SSMM robots is often represented using simple kinematic models neglecting the robot inertias [4], or using dynamic models that assume the mobile base moves on a two-dimensional (2-D) planar surface with wheels that are in permanent contact with the floor [4], the common 3-degree of freedom (DOF) planar base model imposes strong constraints which do not allow us to model the effects of loss of traction or disturbances propagated from the wheel–ground interaction to the arm in the case of field SSMM robots moving in rough terrains. Thus, this paper’s contribution focuses on the development of a general and complete dynamic model that considers a so-called floating base with 6-DOF and nonpermanent wheel–ground contacts.

By general it is meant that the model is applicable to any SSMM configuration, i.e., the model can be employed to describe the dynamic behavior of SSMMs with two or more wheels on each side or tracks approximated as a finite set of contact points (CPs) that are restricted to move along the shape of the track or on a set of small rollers. For detailed track modeling, i.e., considering tracks as assemblies of plates or continuous flexible rubber treads, models must be built using large sets differential algebraic equations and solved with numerical tools and methods such as the one presented in [6]. In contrast, one of the goals of this paper is to obtain the explicit symbolic expressions for the dynamic model, and although SSMM dynamic model developed here considers a single manipulator, the modeling procedure can handle the addition of more arms.

This work extends our preliminary results in [7], presenting here the derivation of the forward dynamics equations for SSMMs using spatial vector algebra [8]. The model does not constrain the mobile base to a 2-D plane like the dynamic models in previous works [9]–[11]. The utility and validity of the model developed is shown by means of experiments with a small MM with a 3-DOF arm and with an industrial skid-steer loader, widely employed in construction and mining. The model equations, the simulation model, and the experiment’s measurements have been made publicly available.¹ It should be noted that some of the existing tools to simulate the dynamics of multi-body systems can deliver solutions with high physical fidelity with different capabilities that are not in the scope of the model proposed here, such as the ability to simulate systems involving closed kinematic chains, flexible bodies, and the interactions of multiple particles; see survey [12] and references therein. However, these tools do not provide explicit symbolic expressions

¹<http://ral.ing.puc.cl/ssmm.htm>

for forward dynamics, which is the focus of this paper's contribution. On the other hand, the existing multibody dynamics simulation tools rely on approaches that demand more computational time when applied to kinematic trees like a SSMM because the forward dynamics problem is solved using Lagrange multiplier methods applied to model formulations expressed in terms of differential algebraic equations using maximal coordinates (see [12, Appendix A], [6]).

The paper is organized as follows. First, a brief review of the related work is presented in Section II. Section III presents the development of the forward dynamic equations for a SSMM whose geometric description is general. The third section also introduces the model of ground reaction forces acting on the wheels CPs. Section IV discusses the numerical simulations and an example application of the model in the evaluation of end-effector motion controllers in the presence of ground disturbances and obstacles. Section V compares results between the simulated dynamics and the measurements obtained using a skid-steer loader. Section VI discuss the main conclusions of this paper.

II. RELATED WORK

There exist several models and control strategies of varying complexity for mobile manipulators (MMs) with standard locomotion topologies (not of the skid-steer type) that have been developed over the last years. For a detailed treatment of MMs modeling and control, readers are referred to [4]. Some MM models consider Ackermann steering and geometry [13], [14], while other consider differential-drive schemes with caster wheels [10], [15], [16]. In general, these MM models separate the vehicle dynamics from that of the manipulator and do not consider the effects of the moving arm on the trajectory of the base. This is usually because the arm mass and velocity are assumed to be negligible, or because they focus on problems of redundancy resolution and trajectory planning. Regarding skid-steer platforms, models often consider only the kinematic equations [17] and treat them as differential drive mobile base [18], neglecting lateral slippage of the CPs that lie on the track or wheels outside of the instantaneous turning circle. This slippage is not the same as the one produced by centrifugal accelerations when the vehicle turns at high speeds, but it affects the steering and guidance of the machine nevertheless. Few authors take into account longitudinal slipping [19] or more detailed lateral skidding models [11], [20], but with simplified ground force reactions that assume the platform is restricted to move on a 2-D plane with constant ground contacts.

Concerning the modeling of wheel-ground interactions, this paper employs a soft contact model in which the wheels or tracks are treated as a rigid body with CPs interacting with a deformable ground surface that reacts according to the Hunt-Crossley spring-damper nonlinear contact force model, but modified according to the recent results in [21]. Other contact force models employ the classic terramechanical models by Bekker, Wong-Reece and their extensions [22]. The classic terramechanical models have been employed in robotics to develop terramechanical parameter estimators [23] and wheel slippage and sinkage detectors [24]. An in-depth review of the

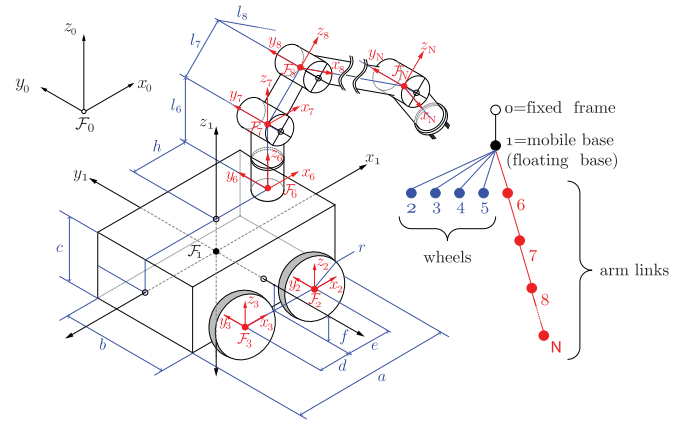


Fig. 1. Skid-steer mobile manipulator model and kinematic tree.

existing contact models can be found in the recent work by Azimi *et al.* [25].

The work by Liu *et al.* [9] is the first one to formulate a dynamic model for SSMMs. The model considers traction and skidding forces, as well as shifts in the robot's center of mass (COM) due to changes in arm position. The SSMM model developed in [9] provides a significant improvement in the representation of the motion behavior of SSMMs, since most of the existing literature only considers kinematic models [4]. However, a drawback of the SSMM model proposed in [9] is that it simplifies the propagation of ground interaction forces up to the arm because it assumes the mobile base is in permanent contact with the ground and restricted to general planar motion. This motivates the development of the SSMM model proposed here, which allows us to represent the effects of loss of traction and the disturbance of uneven terrains because it considers nonpermanent wheel contacts and 6-DOF floating mobile base.

III. SKID-STEER MOBILE MANIPULATOR MODEL

The forward dynamics equations of a general SSMM with a 6-DOF floating base and nonpermanent contacts is developed here using the Articulated Body Algorithm (ABA) [8]. To this end, the geometric description is presented first using the spatial algebra notation and formalism [8], i.e., here *spatial* inertia tensors, transformations, force, and motion vectors are written in upright bold letters (e.g., $\mathbf{I}$, $\mathbf{X}$, $\mathbf{f}$, $\mathbf{v}$), while regular 3-D inertia tensors, transformation matrices, and vectors are denoted using bold italic letters (e.g., $\mathbf{I}$, $\mathbf{E}$, $\mathbf{f}$, $\mathbf{v}$). Scalar quantities are denoted by nonbold italic fonts. The third part of this section presents the wheel-ground interaction model.

A. Geometric Description of a SSMM

The first step is to define the joints and reference frames for the mobile base, its wheels, and the arm links as shown in Fig. 1, starting with the Cartesian coordinate frame $\mathcal{F}_0$ that serves as global inertial frame or virtual fixed base. The frame $\mathcal{F}_1$ is fixed at the COM of the mobile base (body 1), which is treated as a body connected to the fixed base (body 0) by 6-DOF joint, i.e., the mobile base is a *floating base* allowed to move freely without any kinematic constraints unless the wheels are in contact with the ground and the acting contact forces indirectly

TABLE I
GEOMETRIC AND INERTIAL PARAMETERS FOR THE SSMM

Symbol	Description
<i>Geometric parameters</i>	
$\lambda(i)$	Parent of body i .
a, b, c	Base length, width, and height.
d, e	Distance between the rear/front wheels and the base COM.
f	Distance between the axles plane and the base COM.
h	Distance between the manipulator and the base COM.
$\mathbf{r}_i$	Position of body i relative to its parent body $\lambda(i)$.
r, w	Wheels radius and width.
l_i, σ_i	Length and cross-section radius of the i th link.
$\mathcal{F}_0$	Cartesian coordinate inertial frame.
$\mathcal{F}_i$	Reference frame fixed to body i
<i>Inertial parameters</i>	
$\mathbf{c}_i$	Center of mass location of body i in frame $\mathcal{F}_i$ coordinates.
$\mathbf{I}_i$	Inertia matrix of body i at the body's COM.
m_i	Mass of body i .
g	Gravity acceleration constant.

constrain the motion as discussed later in Section III-C. The wheels are bodies labeled 2, 3, 4, and 5 with corresponding Cartesian frames $\mathcal{F}_i$, $i = 2, 3, 4, 5$, connected to the base by rotary joints as shown in Fig. 1. The manipulator is formed by serially connected bodies with coordinate frames $\mathcal{F}_i$, $i = 6, 7, 8, \dots, N$, where N is the last body of the arm and also the total number of bodies of the SSMM. The first link of the manipulator (body 6) is connected to the mobile base (body 1). The connectivity graph for the N bodies of the SSMM is shown in the right side of Fig. 1. The nodes of the graph represent each body, while the lines connecting the nodes represent the joints of the robot. The joints are numbered in such a way that joint i is the joint connecting body i to its parent. The SSMM is a kinematic tree, hence, its connectivity graph is a topological tree. Adding more arms or wheels to the SSMM will append more branches to the tree in Fig. 1. The process of deriving the model equations remains the same regardless of the number of branches, except for the fact that the procedure has to be repeated for all branches that originate from the mobile base. The geometric and inertial parameters of the model are summarized in Table I, in which COM stands for center of mass.

In order to specify the relative locations of the joints in each body, a pair of coordinate frames is introduced for each joint i that connects body i to its parent $\lambda(i)$. One frame is labeled $\mathcal{F}_i$ and fixed to the body i , while the other is labeled $\mathcal{F}_{\lambda(i),i}$ and fixed to the parent body $\lambda(i)$. To minimize the number of parameters required to describe the relative motion between $\mathcal{F}_i$ and $\mathcal{F}_{\lambda(i),i}$, it is convenient to locate the frames such that both frames coincide when the joint variables are zero and their axes are aligned following a set of rules like the widely employed Denavit and Hartenberg convention to constrain the possible frame locations [8]. Noting that each body i contains a frame $\mathcal{F}_i$ and a variable number of frames $\mathcal{F}_{\lambda(j),j}$, for all j satisfying $\lambda(j) = i$ (with frames $\mathcal{F}_{\lambda(j),j}$ located at the joints of the children bodies j whose parent body is body $\lambda(j) = i$), it is convenient to select frame $\mathcal{F}_i$ as the coordinate system for body i , i.e., $\mathcal{F}_i$ is the frame specifying the body coordinates in terms of which the spatial inertia and motion of body i

will be expressed. Finally, a complete description of the robot geometry is obtained defining two transformations: $\mathbf{X}_T(i)$ and $\mathbf{X}_J(i)$. The transformation $\mathbf{X}_T(i) \stackrel{\text{def}}{=} {}^{\lambda(i),i}\mathbf{X}_{\lambda(i)}$ is the Plücker coordinate transform from body $\lambda(i)$ coordinates in frame $\mathcal{F}_{\lambda(i)}$ to coordinates in the frame $\mathcal{F}_{\lambda(i),i}$ located at joint i , but fixed to body $\lambda(i)$. The coordinate transform $\mathbf{X}_J(i) \stackrel{\text{def}}{=} {}^i\mathbf{X}_{\lambda(i),i}$ is the joint coordinate transform mapping coordinates from frame $\mathcal{F}_{\lambda(i),i}$ to coordinates in the frame $\mathcal{F}_i$ fixed to the children body i of the parent body $\lambda(i)$. With these transformations, it is possible to construct the so-called *link-to-link* transform

$${}^i\mathbf{X}_{\lambda(i)} = \mathbf{X}_J(i)\mathbf{X}_T(i) \quad (1)$$

from the frame $\mathcal{F}_{\lambda(i)}$ coordinates of body $\lambda(i)$ to frame $\mathcal{F}_i$ coordinates of body i . The complete set of transformations $\mathbf{X}_T(i)$, $i = 1, 2, \dots, N$, describes the location of each joint-frame $\mathcal{F}_{i,j}$ relative to its corresponding body-frame $\mathcal{F}_i$ within body i . The body constant geometry data contained in $\mathbf{X}_T(i)$ together with the variable joint coordinate transforms $\mathbf{X}_J(i)$, $i = 1, 2, \dots, N$, and the connectivity data in the parent array $\lambda = [\lambda(1), \lambda(2), \dots, \lambda(N)]$ permit us to completely describe the geometry of the robot and the position of its bodies with respect to the global reference frame $\mathcal{F}_0$. The recursive formula

$${}^i\mathbf{X}_0 = {}^i\mathbf{X}_{\lambda(i)} {}^{\lambda(i)}\mathbf{X}_0, \quad \text{with } \lambda(i) \neq 0 \quad (2)$$

which is based on the link-to-link transformation (1), computes the coordinate transform from the global reference frame $\mathcal{F}_0$ to the body coordinate frame $\mathcal{F}_i$, thus allowing us to solve the forward kinematics. Equations (1) and (2) are also an essential part of the recursive inverse kinematics and forward/inverse dynamics computations with algorithms whose detailed discussion can be found in [8]. The specific transformations required to build the SSMM model are summarized in Table II together with the body and joint variables (see [8] for the detailed definition of the transformations $E_u, xlt, rotx, roty, rotz, rotu$, and products $\times, \times^*$).

It is possible to observe in Table II that mobile base is a free-floating body with 6-DOF (three for orientation and three for translation) as defined by the world-to-base transform ${}^1\mathbf{X}_0 = \mathbf{X}_J(1)\mathbf{X}_T(1) = \mathbf{X}_J(1)$ because $\mathbf{X}_T(1) = \mathbb{I}_{6 \times 6}$ for the mobile base. However, to avoid the singularities of the Euler angles (roll, pitch, yaw), the rotations in $\mathbf{X}_J(1)$ are expressed in term of Euler parameters (unit quaternion) involving a rotation axis (u_x, u_y, u_z) and an angular amount θ that give rise to four unit quaternion parameters (p_0, p_1, p_2, p_3) . Therefore, the description of the state of the first body involves a 13-D vector

$$\mathbf{x} = \underbrace{[p_0, p_1, p_2, p_3]}_{\substack{\text{Orientation} \\ \text{quaternion}}} \underbrace{[p_x, p_y, p_z]}_{\substack{\text{Position} \\ \text{relative to} \\ \mathcal{F}_0}} \underbrace{[\omega_x, \omega_y, \omega_z]}_{\substack{\text{Angular} \\ \text{velocity in } \mathcal{F}_0 \\ \text{coordinates}}} \underbrace{[v_x, v_y, v_z]}_{\substack{\text{Linear velocity} \\ \text{in } \mathcal{F}_0 \\ \text{coordinates}}}.$$

For the remaining 1-DOF joints of the wheels and the manipulator, only the angular position q_i and angular velocity $\dot{q}_i$, $i = 2, 3, \dots, N$ are required to complete the description of the state of the bodies. Thus, for a four-wheeled SSMM with an M DOF manipulator, the full state vector is

TABLE II
SUMMARY OF THE SSMM MODEL PARAMETERS AND SPATIAL VECTOR ALGEBRA ARITHMETIC

Joint- i	1	2, 3, 4, 5	6	7, 8, ..., N
Type	Floating	Rotary $\mathbf{y}_i, i = 2, 3, 4, 5$	Rotary $\mathbf{z}_6$	Rotary $\mathbf{y}_i, i = 7, 8, \dots$
Body- i	1	2, 3, 4, 5	6	7, 8, ..., N
Description	Mobile Base	Wheels	First Arm Link	Remaining Arm Links
Parent Body $\lambda(i)$	0	1	1	$6-(N-1)$
Dimensions (x, y, z)	(a, b, c)	$(2r, w, 2r)$	$(\sigma_6, \sigma_6, l_6)$	$(\sigma_i, \sigma_i, l_i)$
Joint-location				
Transforms $\mathbf{X}_T(i)$	$xlt([0, 0, 0]^T)$	$xlt(\mathbf{r}_i)$	$xlt(\mathbf{r}_6)$	$xlt([0, 0, l_{i-1}]^T)$
Joint				
Transforms $\mathbf{X}_J(i)$	$rotu(\cdot)xlt(\mathbf{E}_u^{-1}[p_x, p_y, p_z]^T)$	$roty(q_i)$	$rotz(q_6)$	$roty(q_i)$
Mass m_i	m_1	$m_2 = m_3 = m_4 = m_5$	m_6	m_i
COM $\mathbf{c}_i$ in $\mathcal{F}_i$ coordinates	$[0, 0, 0]^T$	$[0, 0, 0]^T$	$[0, 0, l_6/2]^T$	$[0, 0, l_i/2]^T$
Body inertia $\mathbf{I}_i$ about COM	$\frac{m_1}{12} \begin{bmatrix} b^2 + c^2 & 0 & 0 \\ 0 & a^2 + c^2 & 0 \\ 0 & 0 & a^2 + b^2 \end{bmatrix}$	$\frac{m_i}{12} \begin{bmatrix} 3r^2 + w^2 & 0 & 0 \\ 0 & 6r^2 & 0 \\ 0 & 0 & 3r^2 + w^2 \end{bmatrix}$	$\frac{m_6}{12} \begin{bmatrix} 3\sigma_6^2 + l_6^2 & 0 & 0 \\ 0 & 3\sigma_6^2 + l_6^2 & 0 \\ 0 & 0 & 6\sigma_6^2 \end{bmatrix}$	$\frac{m_i}{12} \begin{bmatrix} 3\sigma_i^2 + l_i^2 & 0 & 0 \\ 0 & 3\sigma_i^2 + l_i^2 & 0 \\ 0 & 0 & 6\sigma_i^2 \end{bmatrix}$

TABLE III
MOTION AND FORCE VARIABLES FOR THE SSMM

Symbol	Description
<i>3D space motion and force variables</i>	
$\boldsymbol{\omega}_i^j, \mathbf{v}_i^j$	Angular and linear velocity of the body i in frame $\mathcal{F}_j$.
$\boldsymbol{\alpha}_i^j, \mathbf{a}_i^j$	Angular and linear acceleration of the body i in frame $\mathcal{F}_j$.
$\boldsymbol{\tau}_i^j, \mathbf{f}_i^j$	Torque and force acting on body i in frame $\mathcal{F}_j$.
$\mathbf{q}_i, \dot{\mathbf{q}}_i, \ddot{\mathbf{q}}_i$	Angular position, velocity and acceleration of the joint i .
$\boldsymbol{\tau}_i$	Torque applied to the joint i .
<i>Spatial space motion and force variables</i>	
$\mathbf{v}_i^j, \mathbf{a}_i^j$	Spatial velocity and acceleration of body i in frame $\mathcal{F}_j$.
$\mathbf{f}_i^j$	Spatial external force applied to the body i .
${}^B \mathbf{X}_A$	Frame motion transformation from body A to B .
${}^B \mathbf{X}_A^*$	Frame force transformation from body A to B .
$\mathbf{I}_i$	Spatial inertia of the body i in the frame $\mathcal{F}_i$.
$\mathbf{I}_{A/B}$	Inertia propagated from body B to body A .
$\mathbf{I}_i'$	Body i total inertia due to children bodies' inertia.
$\mathbf{c}_i$	Joint i spatial acceleration due to velocity-product terms.
$\mathbf{S}_i$	Joint i motion subspace matrix.

given by $\mathbf{q}_{SSMM} = [\mathbf{x}|q_2 \ q_3 \ \dots \ q_{M+5}|\dot{q}_2 \ \dot{q}_3 \ \dots \ \dot{q}_{M+5}]$, requiring $13 + 4 \times 2 + M \times 2 = 13 + 2 \times (M + 4)$ joint variables.

B. SSMM Forward Dynamics Equations

The spatial velocity, acceleration, force and inertia tensors, and their associated 3-D terms required to obtain the expressions for the forward dynamics of the SSMM described in the previous section are summarized in Table III. The notation $\mathbf{v}_i^j$ is employed to indicate that the vector $\mathbf{v}$ is associated with body i and expressed in frame $\mathcal{F}_j$ coordinates. When a variable of body i is expressed in the coordinates of the same body, i.e., when $j = i$, then the superscript is omitted to simplify the notation, thus $\mathbf{v}_i^i = \mathbf{v}_i$.

It will be assumed here that the spatial velocity, acceleration, and force variables of the base $\mathbf{v}_1^0, \mathbf{a}_1^0$, and $\mathbf{f}_1^0$ in the global frame have been expressed as a velocity, acceleration, and force $\mathbf{v}_1^1 \equiv \mathbf{v}_1, \mathbf{a}_1^1 \equiv \mathbf{a}_1$, and $\mathbf{f}_1^1 \equiv \mathbf{f}_1$ with coordinates in the body frame using the base-to-body transformation 1X_0 that maps

coordinates from the global frame $\mathcal{F}_0$ to the base frame $\mathcal{F}_1$. This assumption permits obtaining simpler expressions for the variables in the body frame, than when body variables are expressed in the global frame. On the other hand, typical inertial sensors provide measurements in the body frame of the base. To this end, the base velocity is simply declared as a spatial velocity vector

$$\mathbf{v}_1 = [\omega_{1x} \ \omega_{1y} \ \omega_{1z} \ v_{1x} \ v_{1y} \ v_{1z}]^T \quad (3)$$

whose angular and translational velocity components are referred to the floating body frame $\mathcal{F}_1$. It is to be noted that for numerical simulation it is more convenient to transform the body frame variables to variables with coordinates in the global frame $\mathcal{F}_0$ as stated in the original ABA formulation of the forward dynamics [8].

The spatial inertia matrices of each body are calculated in terms of each body's inertia $\mathbf{I}_i$, mass m_i , and COM location $\mathbf{c}_i$ expressed in body coordinates using the parameters in Table I and the generalized version of the parallel axis theorem for spatial inertias:

$$\mathbf{I}_i = \begin{bmatrix} \mathbf{I}_i + m_i \mathbf{c}_i \times \mathbf{c}_i \times^T & m_i \mathbf{c}_i \times \\ m_i \mathbf{c}_i \times^T & m_i \mathbf{1}_3 \end{bmatrix}.$$

It is to be noted that because the origin of the coordinate frame $\mathcal{F}_1$ of the floating base coincides with the COM location, i.e., $\mathbf{c}_1 = [0, 0, 0]$, the spatial inertia $\mathbf{I}_1$ is easy to calculate, unlike that of the arm links that have nonzero $\mathbf{c}_i$ vectors, which cause off-diagonal elements to appear in the expressions for $\mathbf{I}_i, i = 2, 3, \dots, N$.

Using the floating base velocity in (3) and the joint velocity $\dot{q}_i$ of each body, the spatial velocities of the bodies can now be calculated as the combined contributions of the velocity of the base (parent body) on the wheels/arm (children bodies) and the corresponding joint velocities according to the equation

$$\mathbf{v}_i = {}^i\mathbf{X}_{\lambda(i)} \mathbf{v}_{\lambda(i)} + \mathbf{S}_i \dot{q}_i, \quad i = 2, 3, \dots, N \quad (4)$$

where ${}^i\mathbf{X}_{\lambda(i)}$ is the transformation matrix from the base frame ($\lambda(i) = 1, i = 2, 3, \dots, N$) to the children bodies' frame, while

$\mathbf{S}_i$ is the joint motion subspace matrix, which characterizes the motion constraint imposed by the joint.

The force due to the velocity-product terms $\mathbf{f}_{ic} = \mathbf{v}_i \times^* \mathbf{I}_i \mathbf{v}_i$ can now be calculated using the spatial velocity and inertia of each body. The forces are propagated from the children to their corresponding parent bodies, i.e., from the arm and wheels to the base in the case of the SSMM. The main external forces on the SSMM platform are the gravity force and the ground normal and tangential reaction forces at the CPs of the wheels on the terrain. Therefore, the external spatial force on the mobile base will be given by an expression of the form

$$\begin{aligned} \mathbf{f}_{1\text{ext}} &= [n_{1\text{ext}x} \ n_{1\text{ext}y} \ n_{1\text{ext}z} \ f_{1\text{ext}x} \ f_{1\text{ext}y} \ f_{1\text{ext}z}]^T \\ &= \underbrace{\left[\sum_{i=2}^5 (\mathbf{r}_i - r\hat{\mathbf{k}}_1) \times \mathbf{f}_{iw}^1 \right]}_{\mathbf{f}_{1w}} + \mathbf{f}_{1\text{grav}} \end{aligned} \quad (5)$$

where $\mathbf{r}_i$ is the position vector of each wheel relative to frame $\mathcal{F}_1$, r is the wheel radius, $\hat{\mathbf{k}}_1 = [0 \ 0 \ 1]^T$ is the unit vector parallel to the $\mathbf{z}_1$ -axis, $\mathbf{f}_{iw}^1$ is the reaction force of each wheel, and $\mathbf{f}_{1\text{grav}} = {}^1\mathbf{X}_0^*[0 \ 0 \ 0 \ 0 \ -m_1g]^T$ is the gravity force.

Since the terrain model is often expressed in the global frame $\mathcal{F}_0$, for numerical simulation purposes it may be more convenient to compute the ground contact forces also in the global frame $\mathcal{F}_0$. In such a situation, the contact forces $\mathbf{f}_{iw}^0$, $i = 2, 3, 4, 5$ must be expressed in the frame $\mathcal{F}_1$ coordinates before they can be added to other forces of the mobile base using the transformation $\mathbf{f}_{iw}^1 = {}^1\mathbf{X}_0^*\mathbf{f}_{iw}^0$, where ${}^j\mathbf{X}_0^*$ is the force transform from frame $\mathcal{F}_0$ to frame $\mathcal{F}_1$ coordinates.

The net force on the children bodies of the base (arm links and wheels) is $\mathbf{f}_i = \mathbf{f}_{ic} - \mathbf{f}_{i\text{ext}}$, $i = 2, 3, \dots, N$, where $\mathbf{f}_{ic}$ is the force due to velocity-product terms and $\mathbf{f}_{i\text{ext}}$ is the external force acting on the arm, which can be defined as in (5) by including a contact and gravity force due to the payload if any. The spatial forces of the wheels and arm links are propagated back to the base and used to compute the net force acting on the base according to $\mathbf{f}_1 = \mathbf{f}_{1c} + \sum_{i=2}^N {}^1\mathbf{X}_i^*\mathbf{f}_i - \mathbf{f}_{1\text{ext}}$, where $\mathbf{f}_{1c}$ is the force due to the velocity-product terms and $\mathbf{f}_{1\text{ext}}$ is the external force on the base introduced in (5).

Before calculating the inertial acceleration of the base, it is necessary to compute the apparent inertia of the wheels and arm links as seen by the base. The apparent inertia of any children body i of the base as seen by the base is given by $\mathbf{I}_{1/i} = \mathbf{I}_i - \mathbf{I}_i \mathbf{S}_i (\mathbf{S}_i^T \mathbf{I}_i \mathbf{S}_i)^{-1} \mathbf{S}_i^T \mathbf{I}_i^T$, $i = 2, 3, \dots, N$. Now the total spatial inertia of the floating base $\mathbf{I}'_1$ that includes all the children bodies and the base's own inertia can be calculated as $\mathbf{I}'_1 = \mathbf{I}_1 - \sum_{i=2}^N \mathbf{I}_{1/i}$. Using the spatial force acting on the base and the total spatial inertia of the base, the inertial acceleration of the base is simply given by

$$\begin{aligned} \mathbf{a}_1 &= \underbrace{(\mathbf{I}'_1)^{-1} \mathbf{f}_{1c}}_{\mathbf{a}_{1c}} + \underbrace{(\mathbf{I}'_1)^{-1} \sum_{i=2}^N {}^1\mathbf{X}_i^* \mathbf{f}_i}_{\mathbf{a}_{1\text{ext}}} \\ &\quad - \underbrace{(\mathbf{I}'_1)^{-1} \sum_{i=2}^N {}^1\mathbf{X}_i^* \mathbf{f}_{i\text{ext}}}_{\mathbf{a}_{1/i\text{ext}}} - \underbrace{(\mathbf{I}'_1)^{-1} \mathbf{f}_{1\text{ext}}}_{\mathbf{a}_{1\text{ext}}}. \end{aligned} \quad (6)$$

In the same manner, the velocity of the base is propagated to each one of its children [see (4)], the acceleration of the base is propagated to each one of the child bodies (wheels and arm links), and then added to the children's acceleration due to the joint acceleration and the velocity-product term (corresponding to the spatial cross product of the child body velocity and joint velocity) to obtain the total child body acceleration according to

$$\mathbf{a}_i = {}^i\mathbf{X}_1 \mathbf{a}_1 + \mathbf{c}_i + \mathbf{S}_i \ddot{q}_i, \quad i = 2, 3, \dots, N$$

where $\mathbf{c}_i = \mathbf{v}_i \times \mathbf{S}_i \dot{q}_i$. The angular acceleration of each child body's rotary joint can now be calculated using its spatial acceleration $\mathbf{a}_i$ as

$$\ddot{q}_i = (\tau_i - \mathbf{S}_i^T \mathbf{f}_i - \mathbf{I}_i \mathbf{S}_i ({}^i\mathbf{X}_1 \mathbf{a}_1 + \mathbf{c}_i)) (\mathbf{S}_i^T \mathbf{I}_i \mathbf{S}_i)^{-1}. \quad (7)$$

Equation (7) shows that the joint acceleration $\ddot{q}_i$, $i = 2, 3, \dots, N$ does not only depend on the applied torque τ_i supplied by the joint actuator, but also on the external forces that propagate from the base to the arm/wheels, as well as the load and possible contact forces on the arm/wheels present in $\mathbf{f}_i$. Equations (6) and (7) complete the forward dynamics of the SSMM.

C. Wheel-Ground Interaction

Most of the existing approaches proposed to model ground-wheel contact forces consider a decomposition into a normal and a traction force [22].

The normal force is the projection of the vehicle weight onto the surface normal, while the traction force is directly related to the applied wheel torque and several complex effects that involve tangential and torsional restitution forces, ground internal deformation forces [22], [25], in addition to the tangential Coulomb friction. The local deformation of bodies that are not really rigid causes the CPs to become contact areas. Despite this, a soft contact model can be implemented as the contact between a point and a compliant surface in which the surface behaves as a first-order massless dynamical system that includes not only the tangential Coulomb friction, but also generates damping and spring-like restitution forces that depend on the position and velocity of the contact point relative to the ground surface. To this end, the surface is described here as a piecewise continuous concatenation of planes $\Pi_{x,y}$ with normal vector $\hat{\mathbf{n}}_{x,y}$ in the global coordinate frame $\mathcal{F}_0$ and a distance $\rho_{x,y}$ to the world origin of $\mathcal{F}_0$ (see Fig. 2). The ground friction, stiffness, and damping coefficients that characterize the forces acting on the CPs are denoted by μ , K , and D , respectively. CPs are points fixed to any of the robot's bodies of the form $\mathbf{p}_k = [x_k \ y_k \ z_k]^T$, with coordinates expressed in the global reference frame $\mathcal{F}_0$, and with absolute velocity $\mathbf{v}_k^0 = [v_{kx}^0 \ v_{ky}^0 \ v_{kz}^0]^T$ in $\mathcal{F}_0$ coordinates. The purpose of the CPs is to determine if a given body is in contact with a surface and provides the location at which the contact forces act while the point remains in contact.

The surface penetration distance δ_k and its rate of change, together with the surface stiffness coefficient K and damping coefficient D are employed to compute the magnitude of the normal reaction force at the ground contact point according to

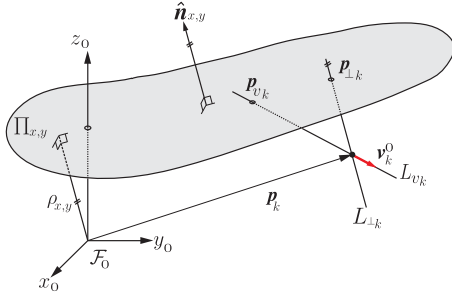


Fig. 2. Projection of contact p_k onto the plane $\Pi_{x,y}$ along the line $L_{\perp k}$ orthogonal to $\Pi_{x,y}$ and the motion direction line L_{v_k} .

the equation

$$N_k = \sqrt{-\delta_k} \left[-K\delta_k - D\dot{\delta}_k \right] \quad (8)$$

where the penetration distance $\delta_k < 0$ for each point in contact with the terrain and the penetration velocity $\dot{\delta}_k$ can be shown to satisfy $\dot{\delta}_k = \hat{n}_{x,y}^T \mathbf{v}_k^0$.

Equation (8) is based on the Hunt–Crossley model of the contact force $N_k = -Kx^n - Dx^p \dot{x}^q$; see [26]. The Hunt–Crossley model is a continuous nonlinear model that describes the compression and restitution forces, as well as the associated energy dissipation of deformable bodies during impact, better than linear models with similar computational requirements. Hence, it is a popular model among the robotics community to describe the contact mechanics; see the review paper [26] and references therein. Many authors employ $n = 3/2$, so that the elastic force term matches that of the Hertzian theory for spheres contacting in static conditions, and $q = 1$ so that the energy dissipation coefficient D of the damping force is easier to estimate. Setting the parameter $p = 0$ simplifies the model further, sacrificing model continuity and some realism, since the dissipation should increase as the contact area increases with the deformation and the development of a plastic region. If the bodies do not undergo significant deformations, it may be possible to assume $p = 0$. Often $p = n = 3/2$ is employed to have a common factor $x^{3/2}$ in both terms [26]. However, recent work in [21] shows that the model accurately fits experimental data when $n = 3/2$, $p = 1/2$, and $q = 1$. Thus, we obtain (8).

Similarly, the tangential compliance of the surface produces a contact tangential reaction force $\mathbf{f}_{tk}$ whose magnitude $f_{tk} \stackrel{\text{def}}{=} \|\mathbf{f}_{tk}\|$ satisfies the Coulomb friction model with coefficient μ and is given by

$$f_{tk} = \begin{cases} \mu N_k, & |\mu N_k| < |f_{\text{stick}_k}| \\ -K_t \epsilon_k - D_t \mathbf{v}_{tk}^0, & |\mu N_k| \geq |f_{\text{stick}_k}| \end{cases} \quad (9)$$

where ϵ_k is the displacement vector of the surface from its equilibrium position in the tangent direction and $\mathbf{v}_{tk}^0$ is the velocity of the surface's tangential deformation. In this model, the magnitude f_{tk} of the tangential reaction force is limited to a so-called sticking force $|f_{\text{stick}_k}|$. The contact point does not slip unless it applies a force on the surface that exceeds the magnitude of the sticking force, i.e., as long as the contact point applies a force that is within the so-called friction cone.

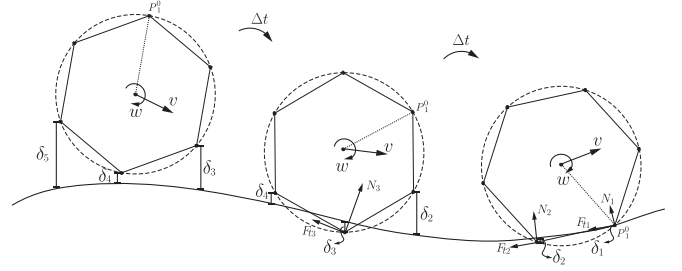


Fig. 3. Wheels contact points transition.

The contact point model for the wheels of the SSMM is implemented using a n -sided polygonal approximation of the wheels, i.e., the ideal circular wheel of radius r with infinite CPs is approximated for practical numerical simulation purposes by a regular polygon inscribed in a circle of radius r with a total of N_{cp} CPs located at each one of the vertices. Fig. 3 shows a wheel with $N_{cp} = 6$ in three different time instants: $t = 0$, $t = \Delta t$, and $t = 2\Delta t$ (from left to right). At each time instant the wheel has a translation and rotational velocity, and each CP has a position defined as p_k , $k = 1, 2, \dots, N_{cp}$, whose velocity is given by $\mathbf{v}_k^0 = \mathbf{v}_{O_i}^0 - r\omega_i^0 \times \hat{n}_k$, where $\mathbf{v}_{O_i}^0$ is the linear velocity of the i th wheel's COM, ω_i^0 is the wheel's angular velocity, r the wheel's radius, and $\hat{n}_k$ is the unitary normal vector of the surface at the contact point.

IV. PATH TRACKING SIMULATION AND EXPERIMENT USING A SSMM WITH A 3-DOF ARM

This section presents an application of the developed SSMM dynamical model in the design of motion controllers for SSMMs whose end-effector must follow some path despite the unevenness of the terrain, such as in harvesting tasks. To this end, a SSMM with a 3-DOF arm is first simulated to obtain the gains that yield the smallest tracking error when its end-effector is required to follow a straight line while the mobile base avoids a virtual obstacle and is driven over a mound. The simulation results are validated using a small SSMM built with a Pioneer P3-AT robot and a Katana 5-DOF arm [see Fig. 4(a)]. The specific values for the geometric and inertial parameters in Table II of this SSMM are summarized in Table IV. Only the first 3-DOF of the arm (waist, shoulder, and elbow) were used in order to keep the experiment simple. It is to be noted that the SSMM is not restricted to planar motion, as shown in Fig. 4(b), which depicts the 3-D view of the path followed by the simulated SSMM. In Fig. 4(b), the green line is the trajectory of the mobile base, the blue line is the reference trajectory of the arm's end-effector, and the red curve is the actual trajectory executed by the arm's end-effector, which is practically overlapping the reference trajectory of the end-effector despite the disturbances caused by the mound and the turning maneuver of the mobile base whose COM's trajectory $\mathbf{c}_1^0 = (p_x, p_y, p_z)$ is shown in Fig. 5. The relatively small tracking error between the end-effector's position $\mathbf{p}_e = (x_e, y_e, z_e)$ and its reference path is clear in Fig. 5.

Employing the SSMMs geometric description provided by the base-to-link transformation in (2), a motion planner

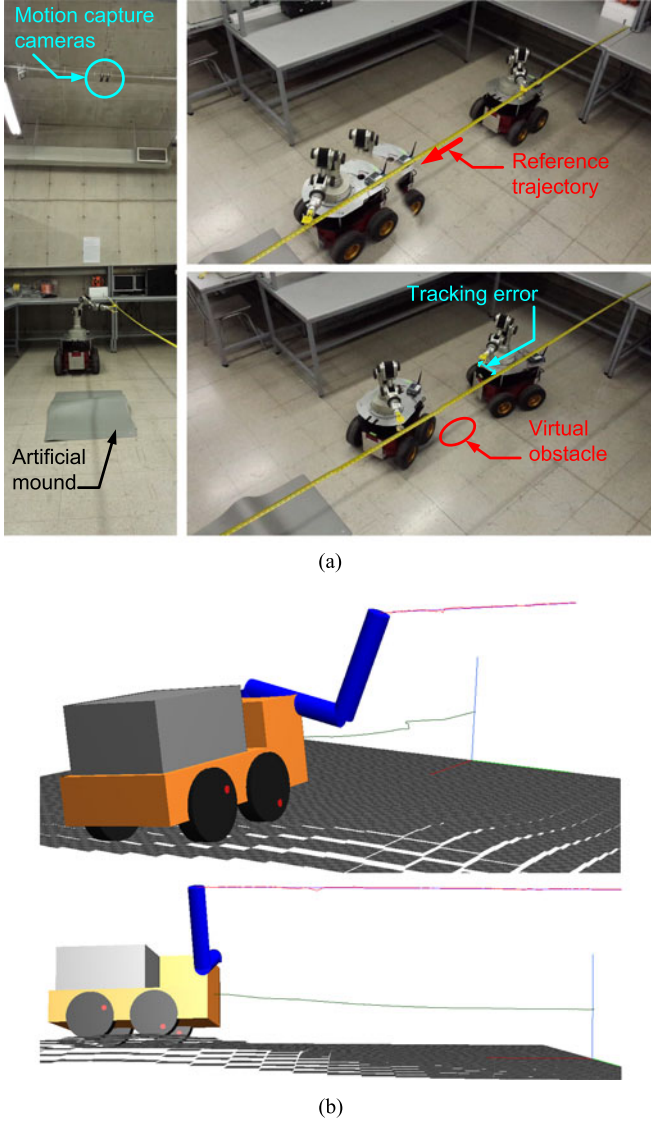


Fig. 4. SSMM with a 3-DOF arm following a reference trajectory with its end-effector while driven over a mound and avoiding a virtual obstacle. (a) Experiment using a P3-AT+Katana Arm. (b) Simulation.

solves the inverse kinematics problem at each simulation step to generate a reference trajectory for the arm joints given the reference trajectory of the end-effector in the task space. Each proportional-integral-derivative (PID) joint controller drives the end-effector to follow the reference trajectory. The development of more advanced computed-torque and predictive control techniques using the developed SSMM model and the measurements of the terrain ahead are part of the ongoing research aimed at decoupling the disturbances of the base on the arm's motion. Nonetheless, the results presented here with the simple PID joint controllers are reasonable because the RMS trajectory tracking error for each of the arm's joints is small. The tuned controllers yield in the simulations RMS errors: $e_{q_6} = 0.7736^\circ$, $e_{q_7} = 0.4641^\circ$, $e_{q_8} = 0.9336^\circ$. Moreover, the RMS end-effector tracking error obtained by averaging ten repetitions of the experiments is 65.3 ± 18.2 mm.

TABLE IV
MODEL PARAMETERS IN SI UNITS FOR THE P3-AT MOBILE BASE WITH A KATANA ARM

Description		Mobile Base	Wheels			Manipulator		
Body		1	2, 3, 4, 5			6	7	8
Dimensions	x	$a :$ 0.508	$2r :$ 0.222	$\epsilon :$ 0.110	0.50	0.50		
	y	$b :$ 0.340	$w :$ 0.100	$\epsilon :$ 0.110	0.50	0.50		
	z	$c :$ 0.126	$2r :$ 0.222	$l :$ 0.393	0.14	0.22		
Mass	$m :$	10	$m :$ 0.5	$m :$ 2.0	1.5	1.5		
Joint			$d :$ 0.142					
Location			$e :$ 0.142	$h :$ 0.100	–	–		
Parameters			$f :$ 0.040					

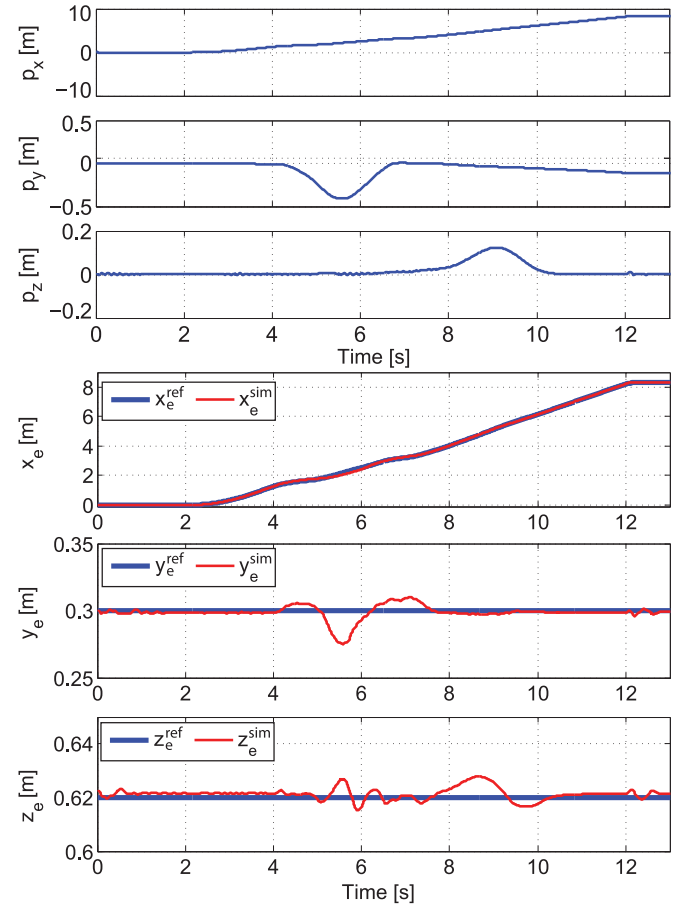


Fig. 5. Simulation results for the position of the mobile base COM $c_1^0 = (p_x, p_y, p_z)$ and end-effector trajectory $p_e = (x_e, y_e, z_e)$ in the global frame $\mathcal{F}_0$ during the execution of an end-effector line-tracking task under mobile base disturbances.

The RMS end-effector tracking error is almost an order of magnitude larger than the simulated error. The larger tracking error obtained with the robot is mainly due to sensor latency and the software interface of the Katana 5-DOF arm which only allows us to send set points without direct manipulation of the currents of each motor to quickly compensate the tracking errors. Parameter uncertainties can also produce a mismatch between how the model represents the loss of traction due to the slippery tiles of the floor in the laboratory and the actual slippage. Nonetheless, a remarkable aspect is that the model



Fig. 6. Compact skid-steer loader at the experiment site with unloaded bucket (left) and loaded bucket (right).

TABLE V
MODEL PARAMETERS IN SI UNITS FOR THE CAT 262C

Description		Mobile Base		Wheels		Manipulator	
Body		1		2, 3, 4, 5		6	
Dimensions	x	$a :$	3	$2r :$	0.9	$\epsilon :$	0.15
	y	$b :$	1.6	$w :$	0.25	$\epsilon :$	0.15
	z	$c :$	1.2	$2r :$	0.9	$l :$	3.3
Mass	$m :$	2389	$m :$	47.7	$m :$	1034	
Joint			$d :$	0.2			
Location			$e :$	0.1	$h :$	-1.2	
Parameters			$f :$	0.5			

facilitated the correct tuning of the controllers, allowing us to reduce the development time and predict the behavior of the SSMM within the margin of error introduced by the hardware and sensors. Further improvements should be possible using a hardware setup with lower latency, as this degrades the performance of the control loops.

V. SIMULATION AND EXPERIMENTAL VALIDATION USING AND INDUSTRIAL SKID-STEER LOADER

The forward dynamics equations developed in Section III for a general SSMM are employed here to describe the motion of an industrial Cat 262C compact skid-steer loader (see Fig. 6) and compared to measurements obtained using an inertial measurement unit (IMU) and an real-time kinematic global positioning system (RTK GPS) during some reference maneuvers. The simulated ground-wheel interactions and the disturbances that they induce on the arm of the SSMM are also compared to the measured vibrations of the mobile base. The latter comparison shows that model can also be used to analyze the reactions of the arm controller.

The specific values for the geometric and inertial parameters in Table II of the Cat 262C model are summarized in Table V. It is to be noted that the parameter h in Table V that defines the joint location of the loader arm (frame $\mathcal{F}_6$ in Fig. 1) is negative. This is because the Cat 262C arm is positioned at the rear-end of the machine, opposite to the front location of the arm in the general SSMM model shown in Fig. 1. The terramechanical model parameters for the elastic and damping force in (8) were set to $K = 6.89 \cdot 10^9 \text{ N/m}^{3/2}$ and $D =$

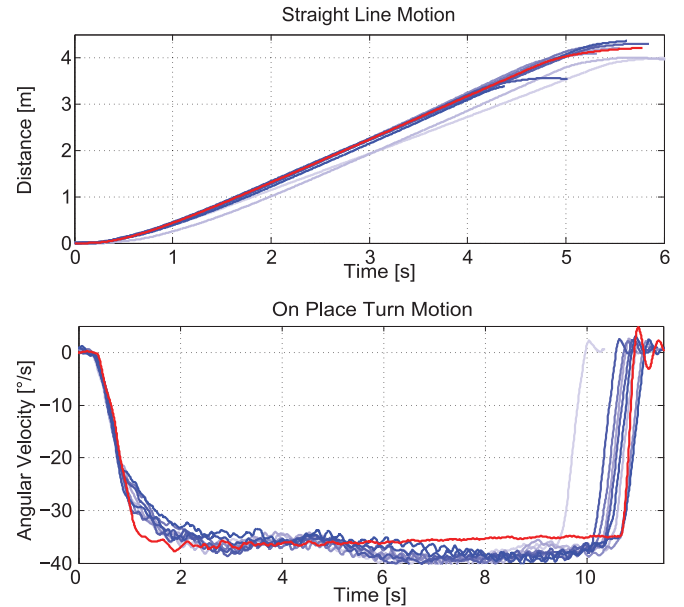


Fig. 7. Experimental results (blue lines) compared to the simulated model (red line) of the SSMM for the straight line motion and in-place turn experiments.

$10^3 \text{ N} \cdot \text{s/m}^{3/2}$, respectively. These values are obtained from standard pavement and rubber wheel stiffness and damping values found in [22] and [27].

A. Comparison of the Simulated Motion and the Experimental Measurements

The experiments consisted of three driving maneuvers: 1) Straight line motion for approximately 4 m; 2) in-place 360° rotations; and 3) a curvilinear path to compare the consistency of the model with the real compact skid-loader. The experiments were carried out keeping the bucket at a constant level and attitude on asphalt pavement. A programmable remote control system was implemented to ensure better repeatability of the motion commands sent to the machine than those performed by a human operator.

The measurements for both straight line motion and in-place rotation show a reasonable agreement with the simulated values. This can be confirmed from Fig. 7, which shows the machine position for the twice integrated acceleration measured in the straight line motion experiment without load (blue lines) and the simulated value for the machine longitudinal position (red line). Similarly, the measurements (blue lines) of the turning rate for the in-place rotation experiment presented in Fig. 7 match the simulated turning rate (red line). The slight mismatch in the curves at the end of each experiment is mainly due to the $\pm 0.5 \text{ s}$ latency of our radio remote control system which affected the duration of each repetition. However, it is possible to observe that slopes of the simulated and measured curves are consistent; thus, the model predicts with fidelity the accelerations and velocities of the machine.

A third experiment in which the compact skid-steer loader steered along a curvilinear path was carried out in order to

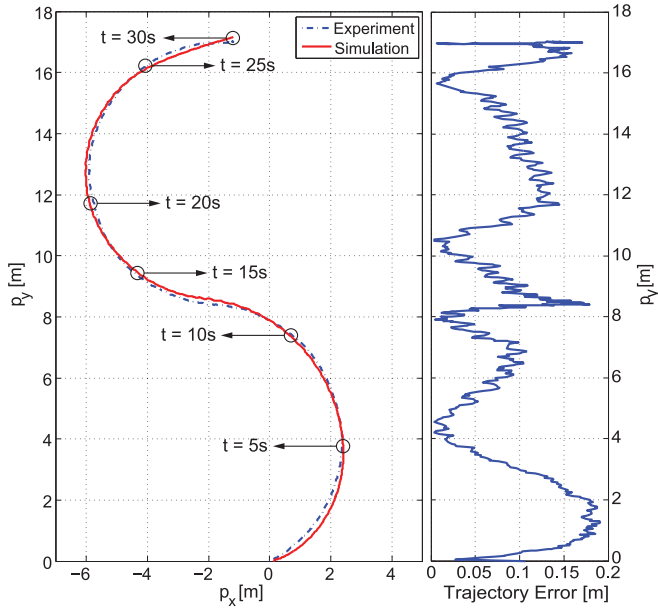


Fig. 8. Comparison of the experimental results (blue dotted line) and the simulation (red line) of the curvilinear path experiment. The absolute error between the measured and the simulated trajectory of the SSMM's base along its path is shown on the right side.

validate the model under more complex maneuvers. In this experiment, the machine was programmed to move with constant longitudinal velocity of 0.82 m/s for 30 s, while applying constant angular velocity command of -0.20 rad/s for the first 15 s, and an angular velocity command of $+0.20$ rad/s for the next 15 s in order to complete an S-shaped trajectory composed of two semicircles of approximately 4 m radius as shown in Fig. 8. The error between the simulated trajectory (red line) and the trajectory measured with the RTK GPS (blue line) shown on the left side of Fig. 8 is very small. In fact, the curve on the right side of Fig. 8 shows that the maximum absolute error is smaller than 0.20 m. The RMS error between measured and the simulated trajectory is below 0.10 m. This result also confirms that the model provides a reasonable estimate of the motion behavior of the real machine.

The simulation results corresponding to the normal forces acting on each of the active CPs of the polygonal wheel approximation of the SSMM are presented in Fig. 9. The upper graph of Fig. 9 shows the normal forces acting on the CPs of the rightfront wheel for the time interval between seconds 2.0 and 2.4 and allows us to see that the forces increase and decrease as each point gradually sinks and leaves the ground while the wheel rotates over the compliant surface. It is possible to observe in the lower graphs of Fig. 9 that the envelope curve of the contact point forces has a maximum value which is different for the left/right and front/rear wheels. This difference is due to the fact that the forces were obtained for the SSMM turning left along an arc segment. Hence, the centrifugal acceleration force of the turning SSMM generates a roll moment about its longitudinal axis x_1 that causes an increase of the reaction force on the wheels that are on the external arc of the turn. Furthermore, since the COM is closer to the rear wheels, the normal reaction

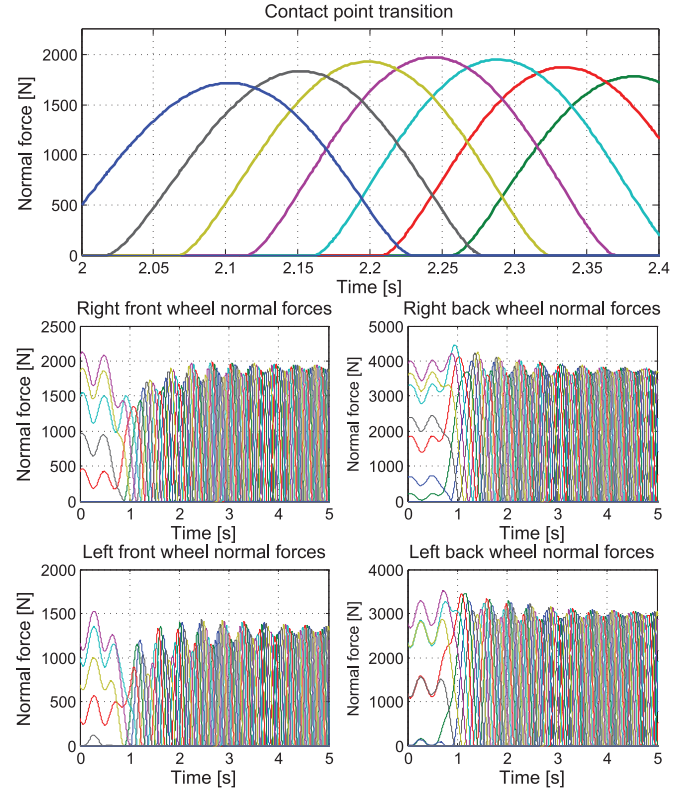


Fig. 9. Evolution of the simulated normal forces acting on the contact points of the wheels of the SSMM performing a left turn maneuver. The upper graph shows a zoomed view of the forces on contact points of the rotating right-front wheel.

force on the rear wheels is almost twice the normal force on the front wheels.

If all the normal forces acting on the active CPs of each of the four wheels are added together, the curves for the total computed normal force shown in Fig. 10 are obtained. Depending on the number of CPs used in the simulation, different vibration frequencies are obtained for the machine moving with the same speed in all cases, as shown in Fig. 10. It is to be noted that regardless of the number of CPs, the addition of the normal forces acting on each wheel produces an average total normal force that is equal to 35429.1 N, which is consistent with the total weight of the skid-steer loader $W_{\text{tot}} = m_{\text{tot}} \cdot g = 35451.4$ N computed using the total mass $m_{\text{tot}} = m_{\text{body}} + m_{\text{arm}} + 4m_{\text{wheel}} = 3613.8$ kg and the gravity acceleration constant $g = 9.81$ m/s². The error between the theoretical normal force and the one obtained from the simulations is less than 0.3% for a number of CPs $N_{\text{cp}} = 32$, and less than 0.1% for $N_{\text{cp}} = 128$. This confirms the validity of the contact point model as it provides a reasonable description of the wheel-ground interaction forces. Furthermore, it is possible to show that if the machine moves with a longitudinal velocity v and its wheels have N_{cp} CPs, the bumping frequency of the CPs against the ground is given by $(vN_{\text{cp}})/(2\pi r)$, then for the simulated skid-steer loader, with v approximately 1 m/s and $r = 0.45$ m, the bumping frequencies can be calculated to be 11.32, 22.64, and 45.28 Hz, for $N_{\text{cp}} = 32, 64, 128$, respectively. Analyzing the FFT computed for the total normal

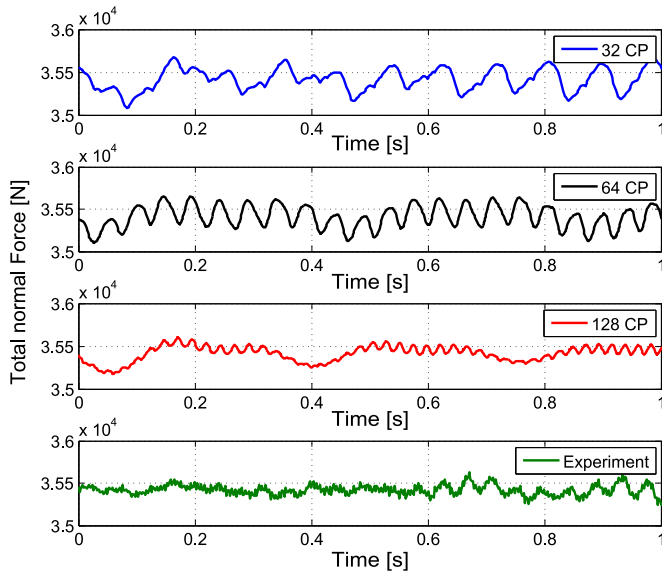


Fig. 10. Total normal force on the SSMM computed from the sum of the normal forces acting on all the active contact points of each of the four wheels obtained through the simulations and the experiment.

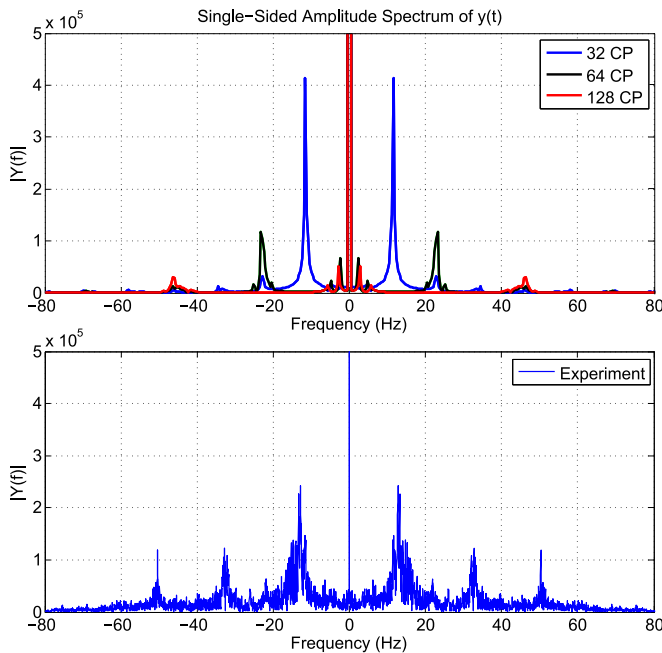


Fig. 11. FFT of the total normal forces acting on the contact points of the wheels, for $p = 32, 64, 128$ contact points models (upper graph) and the experiment (lower graph).

force (see Fig. 11), the peak amplitudes occur at frequencies 11.7, 23.4, and 46.8 Hz, as expected. Comparing the vibrations measured with the IMU, it was observed that the real machine has a similar response to the one produced by the model with 32 CPs per wheel. It is also possible to see in Fig. 11 that the vibrations magnitude decreases with the number of CPs, except for a low frequency oscillation at 3 Hz that is produced by the arm controller, because it is a simple PID that cannot filter the disturbances propagated from the wheels to the base and up to the arm. The coupling between the base and the arm

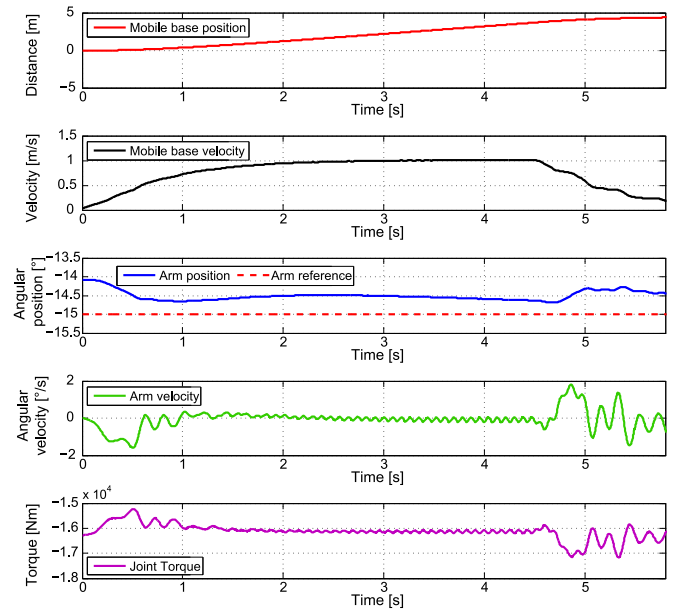


Fig. 12. Simulation results for the closed-loop response of the SSMM's arm when the mobile base follows a trapezoidal velocity profile.

causes the arm vibrations to be propagated back to the wheels, which experiment a minor oscillation in the normal forces that is not caused by the frequency with which the CPs touch the ground.

B. Base-Arm Interaction Simulation

The closed-loop response of the arm when the base is commanded to move according to a trapezoidal velocity profile is shown in Fig. 12. It is possible to observe that the arm initially starting at a position of -14° moves down to -14.5° when the base is accelerated until the speed of the base levels to a value of 1 m/s. The opposite effect occurs on the arm when the base decelerates, as can be seen in the third graph of Fig. 12. The transient perturbations occurring during acceleration and deceleration of the base are more notorious in the fourth graph of Fig. 12, which shows the angular velocity of the arm joint. It can be observed that there is a permanent oscillation in the angular velocity of the arm joint that is caused by the oscillation frequency of the normal forces acting on the CPs that become active as the wheels rotate.

VI. CONCLUSION

The forward dynamics equations of motion for a generic SSMM were developed using the spatial vector algebra approach and the ABA. Unlike the existing models for MMs, the mobile platform is not restricted to 2-D planar motion. The model developed considers the mobile platform as a 6-DOF floating base with nonpermanent ground contacts, i.e., it allows wheels to partially or fully lose traction. The model also considers deformable contacts between the wheels and the ground, thus allows us to compute the effective terrain normal reaction and traction forces acting on each active contact point. The model can be extended to more complex SSMMs having multiple arms with several DOF and more wheels with a user-defined number of CPs.

Although a mathematical model is never a perfect representation of a physical process, the information it provides is accurate to the extent the mathematical model is accurate [28, Ch. 5.10, p. 1003]. The fact that the final outcome of any controller design process is very much dependent on the model chosen and its accuracy is acknowledged even in the robust control literature, which proposes techniques to overcome model uncertainty and process disturbances; see for example [29, Ch. 2, p. 18]. Thus, this paper's contribution focuses on the development of an SSMM model that provides a better representation of the SSMM's dynamics. Two experiments were conducted to assess the utility and validity of the SSMM model. In the first one, the model was employed to describe the motion of a SSMM with a 3-DOF arm whose end-effector was required to follow a straight path despite some disturbances to the mobile base. The second experiment involved an industrial compact skid-steer loader that can be regarded as a SSMM with a 2-DOF planar arm. In both cases, the consistency between experimental and simulated results demonstrate that the model can describe the motion of SSMMs with reasonable physical accuracy suitable for studying the arm-base and the wheel-ground interactions. This feature of the proposed model is very important, especially in field robotics, because it is necessary for the design of decoupling controllers that can reduce the propagation of ground disturbances to the arm. The development of such controllers is part of the ongoing research for teleoperated and semiautonomous underground mining vehicles. Currently, these machines cannot fill their buckets to their maximum capacity because unfiltered vibrations cause them to lose material affecting the operation of other machines. Both the model and experimental data collected during the field tests have been made publicly available at <http://ral.ing.puc.cl/ssmm.htm>.

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